

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA1612

COURSE CODE: Data Warehousing and Data Mining

Q1. Dataset: Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12,000	65
P2	55	25,000	40
P3	42	20,000	75
P4	30	11,000	70
P5	60	28,000	35

SSE Values

K	SSE
1	600
2	350
3	220
4	200
5	190

Answer: The SSE decreases as the number of clusters increases: 600 at K=1, 350 at K=2, 220 at K=3, 200 at K=4, and 190 at K=5. The largest improvement occurs up to K=3. After K=3, the reduction becomes much smaller: only 20 from K=3 to K=4 and 10 from K=4 to K=5. Therefore, the elbow point is approximately K=3, making three clusters the most suitable choice.

With three clusters, patients can be segmented into meaningful groups based on medical expense and health risk. For example, the organization may identify lower-expense/high-risk patients, higher-expense/lower-risk patients, and intermediate or mixed-risk patients. This segmentation supports targeted healthcare planning, resource allocation, preventive care, and personalized intervention. Choosing too few clusters may combine patients with substantially different needs, while too many clusters may create unnecessary and less interpretable groups.

Q2. Dataset: E-Commerce Order Analysis

Order ID	Product Category	Order Value (₹)
O1	Electronics	55,000
O2	Clothing	25,000
O3	Electronics	57,000
O4	Furniture	30,000
O5	Clothing	27,000

Answer: In hierarchical clustering, orders that are most similar are merged first. O1 and O3 are very similar because both belong to Electronics and have high order values of ₹55,000 and ₹57,000. O2 and O5 are also highly similar because both are Clothing orders with values of ₹25,000 and ₹27,000. O4 is a Furniture order of ₹30,000 and is comparatively different from the two strong pairs.

A practical interpretation of the dendrogram would therefore be three clusters: (1) high-value Electronics: O1 and O3; (2) Clothing: O2 and O5; and (3) Furniture: O4. The exact cut level should be selected from the dendrogram by locating a large vertical separation between merges. These clusters help the retailer understand category-specific purchasing patterns, identify high-value product segments, design category-focused promotions, and compare customer/order value behavior.

Q3. Dataset: Mobile App User Behavior

User	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
U1	120	18	2,000
U2	60	5	5,000
U3	140	22	1,500
U4	50	4	6,000
U5	130	20	1,800

Answer: Probability-based or soft clustering assigns each user a membership probability to every cluster instead of forcing the user into exactly one group. From the data, U1, U3, and U5 have high usage time and high login frequency with comparatively lower purchases, so they would likely have high membership in an active-user cluster. U2 and U4 have lower usage and login frequency but higher in-app purchases, so they would likely have high membership in a low-activity/high-purchase cluster.

For example, a user near a cluster boundary could receive memberships such as 0.70 for Cluster A and 0.30 for Cluster B, rather than being assigned only to Cluster A. The actual probabilities must be calculated by the chosen soft-clustering algorithm and cannot be determined from the dataset alone.

Soft clustering is advantageous because real customer behavior is often mixed. A user may be both highly active and a moderate purchaser, or may gradually move from one behavior segment to another. Membership probabilities capture this uncertainty and allow more flexible recommendations, marketing strategies, and user profiling than hard clustering.

Q4. Dataset: Agricultural Crop Monitoring

Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Answer: Before normalization, rainfall is measured in values around 280–520, while soil fertility is only 3–9 and yield is around 65–90. Distance-based clustering can therefore give greater influence to rainfall because it has a larger numerical scale. The resulting clusters may be driven mainly by rainfall rather than by the combined agricultural characteristics.

After normalization, the variables are placed on comparable scales. Consequently, rainfall, soil fertility, and crop yield contribute more fairly to the distance calculation. Farms with similar overall agricultural conditions are more likely to be grouped together. For example, F1 and F4 have high fertility and high yields with relatively lower rainfall, whereas F2 and F5 have high rainfall, lower fertility, and lower yields; F3 represents an intermediate condition.

Normalization improves reliability because no single feature dominates merely due to its units or magnitude. It is especially important for K-Means, hierarchical clustering, and other distance-based methods. The final clusters can support irrigation planning, soil management, crop selection, and yield improvement decisions.

Q5. Dataset: Employee Performance Evaluation

Employee	Productivity Score	Attendance (%)	Skill Assessment Score
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model	Silhouette Score
K=2	0.48
K=3	0.68
K=4	0.52

Answer: The Silhouette Score measures how well each observation fits its own cluster compared with other clusters. A higher value generally indicates better-defined and better-separated clusters. The scores are 0.48 for K=2, 0.68 for K=3, and 0.52 for K=4. Since K=3 has the highest score of 0.68, the most suitable model is K=3.

Using three clusters allows the organization to distinguish employee performance groups more effectively. The clusters could represent high-performing employees, moderate performers, and employees requiring additional support, depending on the actual cluster centroids. Compared with K=2, K=3 provides more detail, while K=4 gives a lower silhouette score and may create less clearly separated groups.

The selected K=3 model can support training plans, recognition programs, workforce development, and performance improvement. However, clustering should be used as a decision-support tool rather than the sole basis for employee evaluation; managerial context and qualitative performance information should also be considered.

Overall Conclusion

The five datasets demonstrate different aspects of clustering interpretation. The hospital problem uses the elbow method to select K=3, the e-commerce problem uses hierarchical relationships visible in a dendrogram, the mobile-app problem illustrates the flexibility of soft clustering, the agricultural problem demonstrates the importance of normalization, and the employee problem uses the Silhouette Score to

select $K=3$. These techniques show that effective clustering requires both numerical evidence and meaningful interpretation of the underlying data.

Instructions

The following five data interpretation questions demonstrate how clustering algorithms can be applied to real-world datasets. Each question includes the data context, interpretation, analysis, application of clustering concepts, conclusion, and a suitable visualization.

Q1. Customer Spending Data

Data Provided / Context: A retail dataset contains annual income and annual spending scores for six customers: A(25,20), B(30,25), C(45,70), D(50,75), E(70,30), and F(75,35). The first value represents income and the second represents spending score.

Analysis and Interpretation: Customers A and B have relatively low income and low spending. Customers C and D have medium-to-high income with high spending, while E and F have high income but comparatively low spending. Therefore, the data shows three natural behavioral groups.

Application of Clustering Concepts: K-Means can be applied with $K=3$. Cluster 1 can represent low-income/low-spending customers, Cluster 2 can represent high-spending customers, and Cluster 3 can represent high-income but lower-spending customers. The resulting groups can be used for targeted marketing.

Conclusion: Customer segmentation helps the retailer design different offers for different customer groups instead of using one marketing strategy for everyone.

Recommended Visualization: A scatter plot of income versus spending with different symbols for each cluster, together with a cluster-centroid plot.

Q2. Mall Customer Segmentation

Data Provided / Context: Suppose a mall dataset contains customers categorized by annual income and spending score. The observed groups show customers with low income/low spending, low income/high spending, high income/low spending, and high income/high spending.

Analysis and Interpretation: The two variables reveal that income alone does not determine spending behavior. Customers with similar income can have very different spending scores. This indicates that segmentation based on both variables is more informative.

Application of Clustering Concepts: K-Means is suitable because the variables are numerical and the objective is to divide customers into distinct groups. The Elbow Method can be used to select the appropriate number of clusters, and the Silhouette Score can evaluate cluster quality.

Conclusion: The clustering result can identify premium customers, potential customers, low-value customers, and customers who spend more despite having lower income. This supports personalized promotions.

Recommended Visualization: Use an Elbow curve for selecting K and a two-dimensional scatter plot showing the final customer clusters.

Q3. Traffic Accident Data

Data Provided / Context: A city transport department records accident locations using latitude and longitude. Most accidents occur in a few dense areas, while some accidents are isolated across the city.

Analysis and Interpretation: The presence of dense geographical groups and isolated observations indicates that the data is not necessarily spherical and contains noise. A density-based approach is therefore more appropriate than a simple centroid-based method.

Application of Clustering Concepts: DBSCAN can group geographically dense accident locations without requiring the number of clusters in advance. The eps parameter controls the neighborhood radius and MinPts defines the minimum number of points needed to form a dense region.

Conclusion: DBSCAN can identify accident hotspots and classify isolated accidents as noise. Authorities can use these results to prioritize traffic signals, road improvements, speed monitoring, and emergency services.

Recommended Visualization: Plot the accident locations on a city map, using different cluster markers and highlighting noise points.

Q4. Manufacturing Sensor Data

Data Provided / Context: A manufacturing plant records temperature, vibration, pressure, and power consumption from machines. Most observations represent normal operation, while a small number show unusual combinations of sensor values.

Analysis and Interpretation: The unusual observations may indicate abnormal machine behavior. Since fault labels may not be available, unsupervised learning can help identify patterns that differ from normal operating conditions.

Application of Clustering Concepts: DBSCAN is appropriate because it can identify dense normal operating patterns and classify isolated observations as noise. Before clustering, sensor variables should be standardized so that features measured on different scales do not dominate the analysis.

Conclusion: The noise points can be investigated as possible maintenance alerts. Repeated abnormal observations from the same machine may indicate a need for inspection or preventive maintenance.

Recommended Visualization: Use a PCA scatter plot showing normal clusters and noise points, plus a time-series chart for abnormal sensor readings.

Q5. Healthcare Service Utilization

Data Provided / Context: A hospital groups de-identified records using variables such as number of visits, average length of stay, healthcare utilization, and selected numerical measurements. The aim is to understand broad patterns of service usage.

Analysis and Interpretation: Patients may form groups with different utilization patterns, such as frequent short visits, fewer but longer stays, or generally low service usage. Standardizing the variables is important before distance-based clustering.

Application of Clustering Concepts: K-Means can be used for an initial segmentation, while Hierarchical Clustering can be used to examine relationships between groups. Silhouette Score and domain knowledge should be used together to assess the usefulness of the clusters.

Conclusion: The clusters can support hospital resource planning, workflow improvement, and service utilization analysis. They should not be treated as medical diagnoses, and results should be validated by domain experts.

Recommended Visualization: Use a PCA scatter plot, cluster-size bar chart, and heatmap showing the average feature values for each cluster.

Overall Data Interpretation

Question	Dataset / Problem	Algorithm	Main Interpretation	Visualization
Q1	Customer spending	K-Means	Three customer behavior groups	Scatter plot
Q2	Mall customer segmentation	K-Means	Income and spending create distinct segments	Elbow + scatter plot
Q3	Traffic accidents	DBSCAN	Dense geographical accident hotspots	City hotspot map
Q4	Machine sensors	DBSCAN	Normal patterns and abnormal observations	PCA + time series
Q5	Healthcare utilization	K-Means + Hierarchical	Different service-use patterns	PCA + heatmap

Final Conclusion

Data interpretation using clustering helps transform unlabelled data into meaningful groups. K-Means is useful for efficient segmentation when clusters are relatively compact, Hierarchical Clustering provides a useful view of relationships among groups, and DBSCAN is especially suitable for dense spatial data and datasets containing noise or outliers. Effective interpretation should combine numerical evaluation, visualizations, and domain knowledge. Thus, clustering algorithms are highly applicable to customer analytics, transportation, manufacturing, healthcare, and many other real-world applications.